

Dr. Nitesh Chawla, Dr. Barry O’Sullivan, and Dr. Richa Singh
Editors-in-Chief, ACM AI Letters

Re: Resubmission of Manuscript AILET-2025-0018
“An Empirical Study of Deep Reinforcement Learning for Dynamic Scheduling in Satellite QKD Networks”

Dear Editors,

Thank you for the opportunity to resubmit our manuscript, and for the constructive assessments from the Associate Editor and the two reviewers. We were encouraged that Reviewer 1 recommended Minor Revision and commended the honesty of the empirical finding, and we have taken the concerns of Reviewer 2 equally seriously. This new manuscript addresses every point raised. The main improvements are as follows.

1. **The reward function is now fully specified and the comparison is provably fair.**
We give the secure-key channel model, the switching-cost dynamics, and the complete piece-wise reward as explicit equations, and we state that every policy (the learned agent and all baselines) is scored by one identical physical objective, the total secure key. No policy is evaluated against a private or differently weighted reward. This directly resolves the central concern of Reviewer 2.
2. **We added a switching-cost sensitivity analysis** (the main request of Reviewer 1). We retrain a DRL agent for each switching-cost value and show how performance, and the learning–heuristic gap, evolve, including a critical cost beyond which all policies fail.
3. **We added a cost-aware hybrid baseline** (greedy with switching hysteresis), tuned by a margin sweep, and explain why it does not beat plain greedy in this geometry-dominated setting.
4. **We strengthened rigor.** All results are now reported as mean and standard deviation over 20 independent seeds, replacing the original single-seed numbers. We also corrected a one-step off-by-one in the switching-cost model so that a nominal cost of C minutes forfeits exactly C minutes, and retrained every model accordingly.
5. **We clarified the motivation and scope**, framing the task as a network-level scheduling problem, justifying the random lower bound, and adding an explicit Limitations discussion on the single-satellite, single-algorithm setting and on generalization.

A point-by-point response to every comment accompanies this letter. The full source code and trained models are available at <https://github.com/ailabteam/AILET>.

Request for an APC waiver. The corresponding author is based at Danang Architecture University in Viet Nam, which the World Bank classifies as a lower-middle-income economy, and we have no institutional or grant funding available to cover an article processing charge. We therefore respectfully request a full (100%) discretionary APC waiver should the manuscript be accepted. We would be grateful for any consideration the editorial office can extend.

We thank you and the reviewers again for the time and care taken with our work, and we hope the revised manuscript is now suitable for publication in ACM AI Letters.

Sincerely,

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